

Final Report:

Section 1- Introduction:

This project analyzes vehicle data to derive insights and build predictive models. Using supervised learning, we aim to predict vehicular pollution levels to support environmental policies. Additionally, unsupervised learning will uncover hidden patterns in the data to provide deeper understanding and potential new categories.

Supervised Learning:

For the supervised learning task, the focus is on predicting the pollution level of vehicles. Vehicular pollution is a major contributor to environmental degradation, affecting public health and climate change. By predicting pollution levels, we can support regulatory efforts to mitigate these effects and encourage the use of environmentally friendly technologies.

Pollution Level Categories:

- **Low Pollution Level (1):** Environmentally friendly vehicles
- **Medium Pollution Level (2):** Vehicles with moderate emissions
- **High Pollution Level (3):** Vehicles with significant environmental impact

Unsupervised Learning:

In the unsupervised learning phase, the aim is to uncover underlying patterns and categories within the dataset. This process helps to identify relationships and trends that may not be immediately obvious, offering additional insights into vehicle pollution trends.

Section 2- Dataset and Features:

API and Dataset Details:

The dataset was obtained through an open API available on [Data.gov.il](https://data.gov.il). The dataset contains information about vehicles in Israel, including vehicle ID, vehicle type, fuel type, pollution level, manufacturing year, and other relevant details. The dataset contains approximately 25,000 records with 26 columns.

Preprocessing Steps:

Removing Irrelevant Columns

We removed irrelevant columns ('misgeret', '_id', 'mispar_rechev', 'mispar_manoa') as they contained unique values per row, making them non-informative for analysis.

Data Translation: Converting Categorical Values to English

We translated categorical values from Hebrew to English, ensuring consistency and accessibility for downstream processes.

Filtering by Manufacturing Year

Based on the distribution of manufacturing years, vehicles manufactured before 1995 were removed as they represent less than 1% of the data, ensuring relevance and focus.

Vehicle Categorization and Analysis

We categorized vehicles into logical groups using predefined patterns and analyzed their distribution for better interpretability.

Target Variable Transformation

The original pollution levels were coded between 0-15, which was too broad and led to insufficient samples for some levels. To address this, we redefined the pollution levels into three broader categories: 1 (Low Pollution), 2 (Medium Pollution), and 3 (High Pollution).

Splitting Data into Training and Testing Sets

At this stage, we split the dataset into training and testing sets, allocating 80% for training and 20% for testing, ensuring robust model evaluation. Until now, the preprocessing steps were applied to the entire dataset. From this point forward, the training and testing sets are handled separately to avoid data leakage, with the same transformations applied to both sets during model training and evaluation.

Handling Missing Values

Removing 'moed_aliya_lakvish' column

We removed the column 'moed_aliya_lakvish' as it was identical to 'shnat_yitzur', ensuring no redundant data in the dataset.

Handling Missing Values in zmig_kidmi

We filled in missing values in the 'zmig_kidmi' column using values from 'zmig_ahori' due to a 90% match between the two. Remaining missing values were filled with the most frequent value, and 'zmig_ahori' was removed to avoid redundancy.

Handling Missing Values in kinuy_mishari

We addressed missing values in the kinuy_mishari column by analyzing the distribution of unique commercial designations (kinuy_mishari) per model (degem_nm). The analysis revealed that most models have a single dominant designation. Based on this, missing values were filled using the most frequent designation per model. For cases where the model was unknown or missing, the global mode of the column was used as a fallback.

Handling Missing Values in ramat_gimur

We filled missing values in ramat_gimur using the most frequent trim level for each model-year combination (degem_nm and shnat_yitzur). For combinations with no available data, we used the global mode as a fallback.

Handling Missing Values in degem_manoa and tozar_manoa

We analyzed the data and found that 88.61% of models (degem_nm) have a single unique value in degem_manoa, and 88.64% in tozar_manoa. Based on this, we filled missing values in these columns using the most frequent value for each model.

Handling Missing Values in mishkal_kolel

We filled missing values in mishkal_kolel using the median weight for each vehicle_category. Missing values were replaced with the corresponding category median to ensure logical and consistent imputation based on vehicle type.

Handling Missing Values in sug_delek_nm

We addressed missing values in sug_delek_nm by filling them with the most frequent fuel type for each vehicle_category. This approach ensures logical imputation based on vehicle type. For categories with no mode available, the global mode of sug_delek_nm was used as a fallback to maintain consistency.

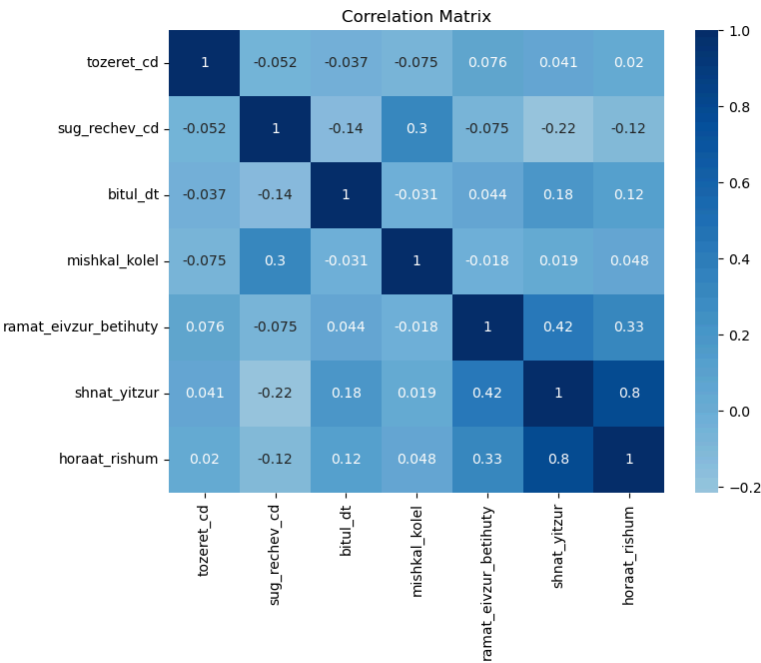
Handling Missing Values with DataImputer

We used the DataImputer class to fill missing numeric values with the median and categorical values with the mode. This ensures any remaining missing values are handled consistently for model readiness.

Handling Outliers

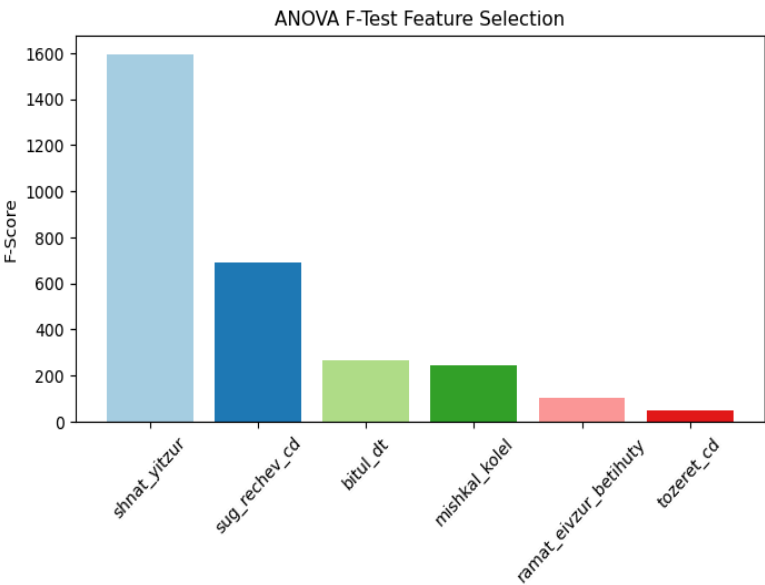
We addressed outliers in mishkal_kolel by defining absolute limits based on vehicle categories and refining bounds using the IQR method. Values outside these bounds were clipped to ensure data consistency while preserving category-specific variability.

Addressing Multicollinearity



We checked for multicollinearity using a correlation heatmap to identify highly correlated features, which can affect linear models like Logistic Regression. A strong correlation (0.8) was found between shnat_yitzur (manufacturing year) and horaat_rishum (registration directive). To reduce redundancy, we removed horaat_rishum.

Feature Selection



Based on the ANOVA F-Test results, tozeret_cd showed a significantly low F-Score, indicating minimal contribution to predicting the target variable. We removed this feature.

Data Transformation

We preprocessed the data by applying one-hot encoding for categorical features, ordinal encoding for ordered features, and standard scaling for numerical features. This ensured consistency and readiness for model training.

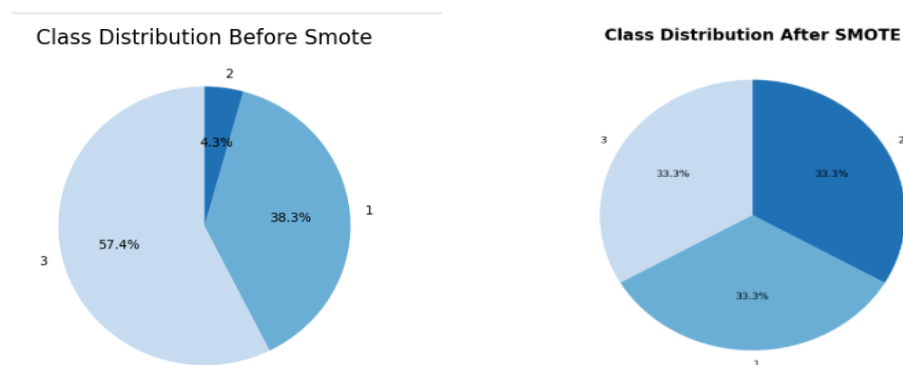
Binary columns: 'degem_manoa', 'degem_nm', 'kinuy_mishari', 'ramat_gimur', 'sug_delek_nm', 'tozar_manoa', 'tozeret_nm', 'tzeva_rechev', 'vehicle_category', 'zmig_kidmi', 'baalut'.

Ordinal columns: 'ramat_eivzur_betihuty'.

Standard columns: 'bitul_dt', 'mishkal_kolel', 'shnat_yitzur', 'sug_rechev_cd'.

Addressing Class Imbalance with SMOTE

To address the class imbalance in the target variable, we applied **SMOTE** (Synthetic Minority Oversampling Technique). This technique generates synthetic samples for underrepresented classes in the training set, effectively balancing the distribution of the target variable. By doing so, we ensured that the model was trained on a more even dataset, which helps improve its ability to recognize and predict the minority classes. This step is particularly important in classification tasks with imbalanced datasets, as it prevents the model from being biased toward the majority class and improves its generalization performance.



Feature Selection Based on Importance

We used feature importance from a trained model (e.g., Random Forest) to retain only features exceeding a specified importance threshold. This reduced dimensionality by filtering out less relevant features, improving model focus and efficiency.

Number of features with importance > 0.001: 170.

Section 3- Methodology:

1. Supervised Learning:

- Logistic Regression:
Used as a baseline for its simplicity and interpretability, helping us understand the linear relationships between features and pollution levels.
- Support Vector Machine (SVM):
Chosen for handling non-linear decision boundaries and high-dimensional spaces, suitable for the complexity of pollution prediction.
- Random Forest:
An ensemble method to capture complex interactions between features and avoid overfitting, providing strong predictive performance.
- K-Nearest Neighbors (KNN):
A simple model based on proximity, useful for capturing non-linear relationships in the dataset.
- Gradient Boosting:
Applied for its ability to improve prediction accuracy by correcting errors from previous models, especially effective for imbalanced datasets and complex interactions.

2. Unsupervised Learning:

Dimensionality Reduction and Clustering:

We initially applied PCA for dimensionality reduction, which helped simplify the dataset by reducing the number of features while retaining important information. However, the clustering results with K-Means using PCA were poor, with a low Silhouette Score (0.07).

To improve the clustering, we used Kernel PCA, which handles non-linear relationships in the data. This transformation allowed us to capture more complex patterns, resulting in better clustering.

Clustering with K-Means:

After dimensionality reduction with Kernel PCA, we applied K-Means with K=2 to partition the data into two clusters. We used the Silhouette Score and Elbow Method to validate the clustering quality and confirm that K=2 was optimal.

Section 4- Experiments and Results:

Parameter Choices:

- **Random Forest:** max_depth=20, n_estimators=100 — Chosen to balance model accuracy and avoid overfitting.
- **SVM:** C=10, kernel='rbf' — Used for handling non-linear decision boundaries, suitable for the complex patterns in pollution prediction.
- **KNN:** n_neighbors=5, weights='distance' — Captured non-linear relationships between data points effectively.
- **Gradient Boosting:** learning_rate=0.1, n_estimators=100 — Applied to improve prediction accuracy and correct errors from previous models.
- **Logistic Regression:** C=10, max_iter=1000 — Chosen as a baseline for simplicity and interpretability, capturing the linear relationship between features and the target variable.

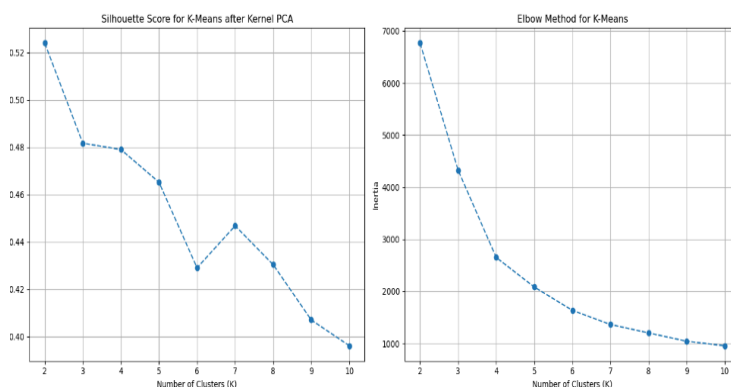
Evaluation Metrics:

- **F1 Score:** Prioritized to balance precision and recall, ensuring a reliable evaluation for class imbalance handled by SMOTE.
- **Accuracy and ROC-AUC:** Used for a comprehensive evaluation of the model's general performance.
- **Precision and Recall:** Ensured the models were correctly detecting pollution levels without false positives or false negatives, especially for the critical categories.

For clustering:

- **Silhouette Score** was used to evaluate how well the clusters were separated and how cohesive each cluster was.
- **Elbow Method** helped determine the optimal number of clusters.

Results:



	Model	Precision	Recall	F1-score	Best Parameters
0	Random Forest	0.975480	0.975279	0.975293	{ 'max_depth': 20, 'n_estimators': 100 }
1	SVM	0.975167	0.973263	0.973829	{ 'C': 10, 'kernel': 'rbf' }
2	KNN	0.969005	0.968024	0.968319	{ 'n_neighbors': 5, 'weights': 'distance' }
4	Gradient Boosting	0.968313	0.961575	0.963510	{ 'learning_rate': 0.1, 'n_estimators': 100 }
3	Logistic Regression	0.962892	0.958081	0.959398	{ 'C': 10, 'max_iter': 1000 }

Best Model: Random Forest and SVM emerged as top performers with F1-scores of 0.975 and 0.973, respectively. Random Forest was ultimately selected due to its robust performance across all pollution categories, minimal overfitting, and better generalization abilities compared to SVM.

Clustering:

The Elbow Method suggested K=4 as the optimal number of clusters, as the inertia (within-cluster sum of squares) showed a noticeable "elbow" at K=4, indicating a point where the improvement in clustering quality begins to level off. However, the Silhouette Score, which measures how similar each point is to its own cluster compared to other clusters, showed the highest value for K=2 (0.524), indicating better cohesion and separation between clusters. After further examination, we tested both K=2 and K=4. The clustering results for K=4 were less distinct, with less meaningful separation between clusters. Therefore, based on the clearer partitioning and higher Silhouette Score for K=2, we selected K=2 as the optimal number of clusters.

Clustering Results: Through the clustering analysis, we observed significant differences between the two clusters. Cluster 0 (the group with older and heavier vehicles) was characterized by a broader range of vehicle weights, and these vehicles had been on the road for extended periods.

In contrast, Cluster 1 consisted primarily of lighter vehicles, most of which were newer (average manufacturing year of 2014), and these vehicles had a relatively shorter time on the road (ranging from 5 to 10 years).

Limitations and Unexpected Outcomes:

1. Class Imbalance:

As mentioned earlier, the class imbalance in the target variable significantly impacted model performance. Certain pollution levels had far fewer samples, which could lead to skewed predictions if unaddressed. Models such as KNN and Logistic Regression, which rely heavily on balanced data, showed limited performance. In contrast, Random Forest and Gradient Boosting showed more resilience.

To address this imbalance, we used SMOTE, which generated synthetic data for the underrepresented classes. This balanced the dataset, improved model stability, and helped ensure more accurate predictions,

2. Pollution Levels and Regulatory Impact: While Cluster 1 (composed of newer, lighter vehicles) exhibited a higher average pollution level (2.85) compared to Cluster 0 (1.03) with older and heavier vehicles, this discrepancy is largely attributed to regulatory changes implemented in 2008. Before this change, most vehicles had relatively low pollution levels (categorized as 1, Low Pollution). After 2008, stricter pollution standards were enforced, which likely explains the higher pollution levels observed in Cluster 1, which predominantly contains newer

vehicles. This finding indicates that pollution levels may not solely be influenced by vehicle age or weight, but rather by changes in national pollution regulations post-2008. This unexpected outcome highlights the importance of understanding regulatory factors when analyzing pollution data.

Section 5 - Conclusion and Discussion

Key Contributions & Implications:

Supervised Learning: Achieved 97.5% accuracy in predicting vehicle pollution levels using Random Forest.

Unsupervised Analysis: Discovered distinct vehicle lifecycles (Cluster 0: 15 years avg, Cluster 1: 9.7 years avg). These findings provide valuable insights for vehicle durability assessment and environmental planning.

Team Contributions:

Itay: Led supervised learning implementation, model optimization.

Gal: Conducted clustering analysis, pattern identification.

Both members collaborated on data preprocessing and results interpretation.

Future Directions:

- Enhance prediction models with additional vehicle characteristics.
- Investigate factors driving lifecycle differences.
- Study economic and environmental implications of vehicle longevity.